

Ricardo (Zhicong) Xie

Product / Hardware Engineering Intern | CAD, Embedded C and Mechatronics

Auckland, New Zealand | +64 29 027 29390 | zxie588@aucklanduni.ac.nz

LinkedIn: <https://www.linkedin.com/in/ricardo-xie-6394643b4> | Portfolio: <https://zxie588-alt.github.io> | GitHub: <https://github.com/zxie588-alt>

NZ student visa | Full-time eligible over scheduled summer break 17 Nov 2026 - late Feb 2027 | No employer visa support needed for this internship | Open NZ-wide

PROFILE

- Mechanical engineering master's student with CAD, enclosure design, embedded C and sensor-system evidence.
- Relevant work: IoT sensor module, registered STM32 weighing firmware, MVHR hardware layout and workshop training.
- Project screenshots and source links: <https://zxie588-alt.github.io>

EDUCATION

University of Auckland - Master of Engineering Studies, Mechanical Engineering | 2026 - Present

Study focus and portfolio work in mechanical, building-services, manufacturing and analysis contexts.

Study record/focus: Completed: Engineering Methods, Computational Fluid Dynamics, Composite Materials, Aerospace Systems Engineering.

Registered S2: Smart Manufacturing, Manufacturing and Industrial Systems.

Huazhong Agricultural University - Bachelor of Engineering, Mechanical Engineering / Mechatronics | 2021 - 2025

Mechanical design, CAD, electrotechnics, sensors, embedded control and mechatronic system integration.

PRACTICAL AND TEAM EXPERIENCE

- Engineering team leadership: led task scoping, workload split and final technical presentation in team design work.
- Workshop practice: prepared AutoCAD/DXF laser-cut geometry and checked cut parts against drawings.
- Hands-on exposure: CNC machining, workholding, cutting fluid, chip formation and 3D printing workflow.
- Lab practice: interpreted force-displacement tests and controlled a robotic arm in training sessions.
- Technical communication: former student news lead; strong project records, photography and presentation visuals.

PROJECT WORK

Ruggedized Outdoor IoT Sensor Module Design Package

SolidWorks, enclosure design, DFM notes, Keil C, testing workflow

- Designed a SolidWorks enclosure with gasket stack-up, screw bosses, PCB envelope and service access.
- Prepared DFM notes for CNC review, 3D printing and assembly tolerance checks.
- Reviewed load paths around corner, strap-lug and screw-boss regions; built Keil C module structure.

Composite Formula SAE Front Wing Analysis

Abaqus/Standard, shell FEA, composite layup, structural checks

- Built an Abaqus shell FEA model with S4R elements, laminate layups and local mesh refinement.
- Converted aero, lateral cornering and 4.1 G bump conditions into structural load cases.
- Checked ply-level stress use; peak utilisation was about 74% under the 4.1 G bump case.

Dynamic Livestock Weighing System - Software Copyright Registered

STM32, HX711, embedded C, Bluetooth/serial communication

- Developed STM32/HX711 weighing firmware with real-time data transfer to a PC interface.
- Received Computer Software Copyright Registration No. 2024SR0141888 for the system.
- Supported dynamic weighing logic for improved signal stability under field-like conditions.

Manufacturing Workshop Training & DXF Preparation

Laser cutting, AutoCAD/DXF, CNC machining exposure, 3D printing workflow

- Prepared AutoCAD/DXF geometry for laser cutting and checked the cut plate against drawing intent.
- Practised CNC workflow: workholding, tool motion, cutting fluid, chip formation and shop safety.
- Reviewed machined gears and shafts for tolerance, edge quality and tool-access constraints.

TECHNICAL SKILLS

- CAD: SolidWorks assemblies, enclosure features, gasket/screw-boss layout and STEP/STL export.
- Embedded: STM32 C, Keil 5, Proteus, HX711, sensors, battery monitor and serial/Bluetooth logic.
- Product support: DFM notes, load-path review, test lists and firmware state-machine records.